The Pharmacopeia of Systems Repair: A Comprehensive Evaluation of Russian Bioregulators, Mitochondrial Modulators, and Epigenetic Agents for Post-Androgen Deprivation Syndromes

1. Introduction: The Epigenetic and Metabolic Stasis of the Post-Finasteride Phenotype

The clinical management of Post-Finasteride Syndrome (PFS) and its related neuro-metabolic dysregulations—often broadly categorized as "Post-Androgen Deprivation Syndrome" (PADS)—has historically been hampered by a reductive focus on serum hormone levels. While the precipitating insult is undoubtedly the pharmacological inhibition of 5-alpha reductase (5AR) enzymes [1], the chronic pathophysiology represents a "systems crash" far more complex than simple hypogonadism. Current systems biology models, specifically those underpinning the "Aurelius Protocol" (AP) [1], characterize this state not as a broken machine, but as an organism trapped in a maladaptive survival program—a persistent **Cell Danger Response (CDR)**.
In this state, the body prioritizes survival over reproduction and anabolic repair. This manifests as mitochondrial hibernation (hypometabolism), epigenetic silencing of androgen receptor (AR) and neurosteroidogenic genes, and a fundamental uncoupling of the neurovascular and gut-immune axes.[1] Standard interventions (TRT, PDE5 inhibitors) often fail because they attempt to force "fuel" into an engine that has been epigenetically locked in "safe mode." The organism perceives an environment of scarcity or toxicity and thus downregulates "expensive" biological processes such as sexual drive, higher-order cognition, and tissue regeneration.
The next frontier of therapeutic intervention lies in **Bioregulation**—the use of signaling peptides and metabolic modulators to epigenetically "reboot" these stalled systems. While Western medicine has focused heavily on small-molecule inhibitors and receptor agonists, a distinct and highly sophisticated tradition of "Bioregulatory Medicine" emerged in the Soviet Union and Eastern Europe, driven by the military's need to protect personnel from extreme stress, radiation, and biological warfare. These agents—specifically **Cytomedins** (tissue-
...n **Peptides**—offer a
...y do not merely agonize or
...omatin level, inducing

...er-known" global
...ng peptides" (BPC-157, TB-
...fractory "invisible symptoms"
...hedonia
...ic stagnation. We categorize
...n into **Phase 3**

(Experimental/Optimization) of the Aurelius architecture, providing a rigorous risk-benefit analysis for the "Captain" and other advanced operators. This analysis integrates genetic predispositions—specifically methylation status, ALDH2 proficiency, and mitochondrial dynamics—to explain the paradoxical inversion of libido and drive, offering a roadmap for reversal that respects the body's homeostatic priority.

1.1 The Theoretical Framework: From Receptor Blockade to Chromatin Remodeling

To understand the utility of Russian bioregulators, one must appreciate the distinction between "pharmacology" and "bioregulation." Standard pharmacology targets the *downstream* effects of protein function (e.g., blocking an enzyme, stimulating a receptor). Bioregulation targets the *upstream* production of those proteins.

Research by Khavinson and colleagues at the St. Petersburg Institute of Bioregulation and Gerontology has demonstrated that short peptides (2-4 amino acids) can penetrate the cell nucleus and bind to specific DNA sequences (promoter regions) in a sequence-specific manner.[3] This binding induces the uncoiling of chromatin (heterochromatin to euchromatin), making silent genes accessible to transcription factors. In the context of PFS, where the 5-alpha reductase gene (*SRD5A2*) and Androgen Receptor (*AR*) may be epigenetically methylated or silenced [1], these peptides offer a theoretical "crowbar" to pry open the genetic machinery that standard hormones cannot reach.

Furthermore, the concept of "Cytomedins"—complexes of polypeptide fractions extracted from young animal tissues—relies on the principle of tissue-specific homology. A peptide extracted from the pineal gland (*Epithalamin*) specifically targets the pineal gland; a peptide from the prostate (*Libidon*) targets the prostate. This allows for a "surgical" restoration of organ function without the systemic side effects of exogenous hormone replacement.

---

2. Module A: The Ophthalmo-Hepatic Bridge and ALDH Restoration

Target: *Retinalamin, Epithalon, and the Retinoid-ALDH Axis*

One of the most enigmatic and debilitating symptoms of PFS is the "functional ALDH deficiency," manifesting as severe alcohol intolerance, brain fog, and visual disturbances (night blindness, "visual snow").[1] The Aurelius Protocol correctly identifies the **Aldehyde Dehydrogenase (ALDH)** enzyme family as a critical bottleneck. ALDH is not only required for detoxifying ethanol and gut-derived acetaldehyde; it is the rate-limiting enzyme in the synthesis of **Retinoic Acid (RA)** from Vitamin A (Retinaldehyde).[5]

The failure of the ALDH axis creates a "toxic feedback loop." High levels of acetaldehyde inhibit mitochondrial function and deplete glutathione. Simultaneously, the lack of Retinoic Acid (a potent transcription factor) prevents the repair of epithelial tissues (gut, skin, eyes) and impairs stem cell differentiation. This explains the "Cluster A" symptoms described in the protocol: night blindness, dry eyes, and "sandy" vision.[1]

2.1 Retinalamin: Beyond Vision to Systemic Retinoid Repair

Retinalamin is a complex of polypeptide fractions extracted from the retina of cattle. While marketed primarily for retinal degenerative diseases (glaucoma, macular degeneration) [7], its relevance to PFS is profound and largely unrecognized in the West.

2.1.1 Mechanism of Action: The ALDH Connection

The retina is one of the most metabolically active tissues in the body and a primary site of retinoid metabolism. Retinalamin has been shown to:

- **Stimulate Retinal Gene Expression:** It regulates the expression of genes involved in the visual cycle and neuroprotection.[7]

- **Modulate ALDH Activity:** Crucially, the retina is rich in **ALDH1A1** and **ALDH3A1** enzymes, which act as "suicide enzymes" absorbing UV radiation and detoxifying lipid peroxidation products (4-HNE, MDA).[6] Research indicates that retinal peptides can upregulate the expression of these specific ALDH isoforms to protect the neurovascular unit.[6]
- **Neuro-Vascular Coupling:** It restores the integrity of the blood-retina barrier, which is structurally homologous to the blood-brain barrier (BBB). By repairing the retinal vasculature, it may exert pleiotropic effects on cerebral microcirculation.

Second-Order Insight: In PFS, the downregulation of 5AR disrupts the androgen-mediated control of ALDH expression.[1] By introducing Retinalamin, we may be providing the specific "code" to re-awaken ALDH expression not just in the eye, but systemically (via the retinoid signaling axis), thereby addressing the "Retinoid Crash" and alcohol sensitivity described in the protocol.[1] The retina acts as a "window" to the brain's metabolic state; repairing retinal ALDH status may signal a broader metabolic "all clear" to the central nervous system.

2.1.2 Integration into Aurelius (Phase 3)

- **Classification: EXPERIMENT / REPAIR**
- **Dosing:** 5–10 mg intramuscularly (IM) daily for 10 days. Repeat every 3-6 months.
- **Synergy:** Combined with **Zinc** and **Vitamin A (Retinol)** (Phase 1 Support) to provide substrate for the upregulated enzymes. Zinc is the obligate cofactor for ALDH; Retinalamin provides the signal to build the enzyme.
- **Expected Outcome:** Improved night vision, reduction in "visual snow," and potentially improved tolerance to aldehydes (less brain fog after meals).

2.2 Epithalon (Epithalamin): The Pineal-Retinal Axis

Epithalon (synthetic tetrapeptide Ala-Glu-Asp-Gly) is the synthetic version of **Epithalamin** (pineal gland extract). It is famous for telomerase activation [10], but its role in the **Pineal-Retinal Axis** is critical for PFS.

2.2.1 Mechanism: Melatonin and Circadian Reset

The pineal gland and retina share a common embryological origin (the "third eye"). Epithalon:

- **Restores Melatonin Synthesis:** It normalizes the synthesis of serotonin and melatonin in the pineal gland, which is often calcified or dysregulated in aging and chronic stress (CDR).[4]
- **Gene Expression:** It induces the expression of transcription factors that regulate antioxidant defenses (SOD, Glutathione Peroxidase).[4]
- **Telomere Elongation:** It activates telomerase in somatic cells, potentially reversing the "accelerated aging" phenotype (Cluster C in Aurelius reports).[12]

Third-Order Insight: Melatonin is not just a sleep hormone; it is a potent antioxidant that specifically protects the mitochondria and the gonads. By restoring endogenous melatonin rhythms via Epithalon (rather than exogenous supplementation), we restore the "night phase" of the circadian cycle, which is the only time the brain glymphatic system clears toxins and neurosteroid receptors reset. This addresses the sleep architecture fragmentation seen in PFS.[1] Furthermore, the pineal gland regulates the sensitivity of the hypothalamus to hormonal feedback; resetting pineal function may help "unstick" the HPG axis.

2.2.2 Pinealon: The Cognitive Sibling

While Epithalon targets the pineal gland's endocrine output, **Pinealon** (synthetic tripeptide Glu-Arg-Glu) is a bioregulator specifically for the brain cortex and neuro-circadian synchronization.[14]

- **Mechanism:** It penetrates the blood-brain barrier and nuclei of neurons, regulating the metabolism of brain cells. It has shown efficacy in correcting circadian rhythm disturbances in shift workers and aging individuals.[14]
- **PFS Application:** For the patient with severe "brain fog" and "day/night reversal," Pinealon may offer a more direct cognitive benefit than Epithalon, acting as a "nootropic for the circadian clock."

---

3. Module B: The Thymic-Neuro Restoration (Immune-Endocrine Reset)

Target: *Thymalin, Thymosin Alpha-1, Vilon, and the Reversal of Immunosenescence*
The Aurelius framework highlights the "Gut-Immune Interface" as a source of chronic inflammation (endotoxemia) that suppresses testosterone.[1] Chronic stress and androgen deprivation lead to premature involution of the thymus gland, resulting in a deficit of T-cells and a failure of immune surveillance. This "immune exhaustion" is a hallmark of the Cell Danger Response.

3.1 Thymalin: The Bioregulator of Immunity

Thymalin is a polypeptide complex extracted from the thymus gland. Unlike **Thymosin Alpha-1** (a single synthetic peptide), Thymalin contains a spectrum of thymic factors.[2]

3.1.1 Mechanism: Hematopoietic Differentiation

- **T-Cell Maturation:** Thymalin stimulates the differentiation of hematopoietic stem cells (HSCs) into functional T-lymphocytes (CD3+, CD4+).[2]
- **Cytokine Modulation:** It normalizes the ratio of pro-inflammatory and anti-inflammatory cytokines, dampening the "cytokine storm" or chronic low-grade inflammation driven by gut endotoxin.[2]
- **Cellular Regeneration:** Evidence suggests it promotes hemostasis and tissue regeneration, likely by modulating the immune response to injury.[3]

Comparative Insight: While **Thymosin Alpha-1 (Zadaxin)** [16] is excellent for acute viral defense (Th1 boosting), **Thymalin** appears to have a broader "geroprotective" and restorative effect on the thymus gland itself. For a PFS patient with "immune exhaustion" or autoimmune-like symptoms (joint pain, food sensitivities), Thymalin offers a deeper "hardware reset" of the immune system.

3.2 Vilon: The Synthetic Thymic Activator

Vilon (Lys-Glu) is the synthetic dipeptide derived from Thymalin.

- **Mechanism:** It induces the expression of interleukin-2 (IL-2) and its receptor, critical for T-cell proliferation.[3] It acts as a chromatin remodeler, specifically uncoiling heterochromatin in lymphocytes.
- **PFS Application:** Vilon is essentially the "active core" of the thymic regeneration signal. It is often used in Russian protocols alongside Epithalon to simultaneously rejuvenate the immune and endocrine systems (the "gold standard" dual revitalization).

3.3 LL-37 (Cathelicidin): The Gut Barrier Architect

LL-37 is the only human cathelicidin, an antimicrobial peptide (AMP) expressed in the colon, skin, and immune cells.[18]

3.3.1 Mechanism: Endotoxin Neutralization and Barrier Repair

- **LPS Neutralization:** LL-37 binds directly to Lipopolysaccharide (LPS/Endotoxin), preventing it from activating Toll-Like Receptor 4 (TLR4). This stops the inflammatory cascade *at the source*.[18] This is critical for the "Endotoxin Brake" on libido described in HFMP.[1]
- **Re-Epithelialization:** It promotes the migration of colonic epithelial cells, physically healing "leaky gut" (intestinal permeability).[20]
- **Microbiome Modulation:** It has broad-spectrum antibiotic activity against pathogens (Candida, SIBO) while sparing commensal bacteria, acting as a "smart antibiotic".[18]

Risk/Reward Analysis: LL-37 is powerful but can be pro-inflammatory or "tumorgenic" in specific contexts (like existing cancer) because it stimulates growth factors (IGF-1R, EGFR).[20] It should be used in short, targeted "kill/repair" cycles, not chronically. It represents a "Phase 3" nuclear option for refractory gut issues.

---

4. Module C: Mitochondrial Ignition and "Hibernation" Breaking

Target: *Actovegin, Meldonium, SS-31, MOTS-c, and the "Turnbuckle" Protocol*
The "Metabolic Hibernation" or Cell Danger Response (CDR) theory posits that PFS cells are stuck in a hypometabolic state, refusing to oxidize fuel efficiently.[1] The goal is to force the mitochondria back "online" through a combination of structural repair and metabolic forcing.

4.1 Actovegin: The "Hypoxia Eraser"

Actovegin is a deproteinized hemodialysate of calf blood. It has been used for decades in Eastern Europe for stroke and diabetic neuropathy.[22]

4.1.1 Mechanism: Oxygen and Glucose Uptake

- **Insulin-Mimetic:** Actovegin stimulates glucose transport (GLUT) into cells independent of insulin. It forces glucose into the "starving" neurons and muscle cells.[23]
- **Oxygen Utilization:** It increases oxygen consumption and uptake by mitochondria, shifting cells away from anaerobic glycolysis (lactate) and back to oxidative phosphorylation (OXPHOS).[22]
- **Neuroprotection:** It reduces PARP activity and NF-kB activation, protecting neurons from apoptosis during metabolic stress.[23]

Application in PFS: Actovegin directly counters the "hypometabolic" phenotype (cold hands, lethargy). It acts as a "spark plug" to reignite energy production in tissues that have down-throttled due to androgen deprivation.

4.2 Meldonium (Mildronate): The Metabolic Shifter

Meldonium is famous as a performance-enhancing drug (banned by WADA), but its therapeutic mechanism is uniquely suited for the "PFS Metabolic Trap".[26]

4.2.1 Mechanism: Carnitine Inhibition and Glucose Preference

- **Inhibits GBB Hydroxylase:** Meldonium inhibits the enzyme that synthesizes carnitine. This lowers intracellular carnitine levels.[26]
- **Blocks Fatty Acid Oxidation (FAO):** Without carnitine, long-chain fatty acids cannot enter the mitochondria. This prevents the accumulation of toxic acylcarnitines (intermediates that cause insulin resistance).[26]

- **Forces Glycolysis:** By blocking fat burning, the cell *must* burn glucose. Glycolysis is more oxygen-efficient than FAO. This protects cells from ischemia and "retrains" the metabolic machinery to use glucose effectively.[29]

Third-Order Insight: In PFS, there is often a "metabolic inflexibility" where cells can neither burn fat nor sugar efficiently. By *blocking* fat (which might be "clogging" the system with toxic intermediates due to stalled beta-oxidation), Meldonium forces the mitochondria to clear the glucose backlog. This aligns with the "Warburg Effect in non-cancerous tissue" described in the user's notes.[1]

4.3 SS-31 (Elamipretide): The Cardiolipin Stabilizer

SS-31 is a tetrapeptide that targets the inner mitochondrial membrane (IMM).[31]

4.3.1 Mechanism: Structural Repair of the ETC

- **Cardiolipin Binding:** SS-31 penetrates the mitochondria and binds specifically to **Cardiolipin**, a phospholipid crucial for the curvature of the cristae and the assembly of the Electron Transport Chain (ETC) supercomplexes.[32]
- **ROS Scavenging:** Unlike general antioxidants (Vitamin C), SS-31 prevents ROS formation *at the source* (electron leak) without blocking physiological ROS signaling.[32]
- **Restores Bioenergetics:** By fixing the physical structure of the IMM, it restores ATP production and membrane potential.[34]

Synergy: SS-31 repairs the *structure* (engine), while Actovegin provides the *fuel/oxygen* (intake), and Meldonium tunes the *fuel mix* (ECU).

4.4 The "Turnbuckle" Mitochondrial Dynamics Protocol

Derived from the Longecity longevity forums, the "Turnbuckle" protocol focuses on manipulating the **fission** and **fusion** cycles of mitochondria to purge dysfunctional units (mitophagy) and stimulate biogenesis.[35]

- **Objective:** To induce a "clean sweep" of the mitochondrial network.
- **Components:**
  - **Fission Agents:** Compounds that promote fragmentation (e.g., **C60** in some interpretations, or specific herbal uncouplers). This isolates damaged mitochondria.
  - **Fusion Agents: Stearic Acid** (C18:0) is identified as a promoter of mitochondrial fusion.[35] Fusion allows mitochondria to mix contents and repair.
  - **Stem Cell Renewal:** The protocol uses **C60 (Buckminsterfullerene)** in olive oil to protect stem cells during this turnover.[35]

Relevance: PFS is often characterized by a "senescent" mitochondrial phenotype—fused, giant, but dysfunctional mitochondria. Inducing fission (to break them up) followed by fusion (to rebuild) is a theoretical way to "de-age" the mitochondrial network. This is highly experimental but aligns with the "Cell Danger Response" reversal strategy.

4.5 MOTS-c: The Mitochondrial Hormone

MOTS-c (Mitochondrial Open Reading Frame of the 12S rRNA-c) is a peptide encoded by the mitochondrial DNA itself, acting as a hormone.[38]

- **Mechanism:** It translocates to the nucleus to regulate nuclear gene expression in response to metabolic stress. It promotes insulin sensitivity, fatty acid oxidation, and prevents diet-induced obesity.[39]
- **Exercise Mimetic:** It activates AMPK and mimics the metabolic effects of exercise.[38] For PFS patients with severe fatigue/exercise intolerance (PEM), MOTS-c can provide the metabolic signal of exercise without the physical stress cost.

5. Module D: The Gonadal "Hardware" Reset

Target: *Testoluten, Libidon, and the Androgen Receptor (AR)*
While TRT provides the hormone, it does not fix the *factory*. The Russian bioregulators **Testoluten** and **Libidon** offer a way to regenerate the testicular and prostatic tissue itself.

5.1 Testoluten: The Testicular Bioregulator

Testoluten is a peptide complex extracted from the testes of young animals.

5.1.1 Mechanism: Leydig Cell Restoration

- **Transcriptional Activation:** It binds to the promoter regions of genes involved in steroidogenesis, specifically **StAR** (Steroidogenic Acute Regulatory protein) and enzymes like CYP11A1.[40]
- **Functional Restoration:** Unlike hCG (which mimics LH and can desensitize receptors), Testoluten rejuvenates the Leydig cells' *capacity* to respond to LH. It increases endogenous testosterone production by restoring the cellular machinery.[40]
- **Sperm Quality:** It improves spermatogenesis and sperm motility, indicating a broad trophism for testicular tissue.

Relevance: For the PFS patient with "normal" T but low drive, Testoluten may restore the *quality* and *pattern* of androgen release (pulsatility), which TRT often flattens.

5.2 Libidon: The Prostate Bioregulator

Libidon is a peptide complex extracted from the prostate gland.

5.2.1 Mechanism: Prostatic Health and 5AR Expression

- **Prostatic Trophism:** It reduces inflammation and improves the functional state of the prostate gland.[42]
- **5-Alpha Reductase Modulation:** The prostate is the primary site of 5-alpha reductase (Type 2). By restoring prostatic cellular health, Libidon may theoretically support the re-expression of 5AR enzymes that were silenced by Finasteride.
- **Symptom Relief:** It is used for Benign Prostatic Hyperplasia (BPH) and prostatitis, reducing urinary symptoms and pelvic discomfort (Cluster D: "Hard Flaccid" connection).[42]

Strategic Use: Restoring the prostate is not just about urinary health; the prostate is a major source of systemic DHT. Healing the prostate via Libidon is a strategy to bring the body's primary "DHT factory" back online.

---

6. Module E: Neuro-Synaptic Reconstruction and Membrane Stabilization

Target: *Cortexin, Cerebrolysin, Dihexa, 9-Me-BC, Emoxypine*
The neurological symptoms of PFS (anhedonia, cognitive fog, loss of libido) are driven by neuroinflammation, synaptic pruning, and membrane instability.

6.1 Cortexin: The Cortical Specific Repair

Cortexin is a complex of polypeptide fractions from the cerebral cortex of cattle/pigs.[43]

6.1.1 Mechanism: Balanced Neuromodulation

- **Neurotrophic Factors:** It contains endogenous BDNF and NGF, stimulating neurite

outgrowth.[44]
- **GABA/Glutamate Balance:** Unlike Cerebrolysin (which is whole-brain), Cortexin has a specific modulatory effect on **GABAergic** and **Glutamatergic** transmission, reducing excitotoxicity while restoring inhibition.[45]
- **Mechanism:** It restores the balance of excitatory and inhibitory amino acids and modulates serotonin/dopamine levels.[46]

Relevance to PFS: The "wired-tired" anxiety and insomnia in PFS are linked to a GABA deficit (Allopregnanolone crash). Cortexin's GABAergic activity makes it superior to Cerebrolysin for this specific phenotype.

6.2 Dihexa: The Synaptogenic Heavy Hitter

Dihexa is a synthetic peptide derived from Angiotensin IV.[47]

6.2.1 Mechanism: HGF/c-Met Activation

- **HGF Mimetic:** It binds to Hepatocyte Growth Factor (HGF) and creates a functional ligand that activates the **c-Met receptor**.[47]
- **Synaptogenesis:** It is claimed to be seven orders of magnitude more potent than BDNF at creating new synapses (spinogenesis).[50]
- **Oral Bioavailability:** Unusually for a peptide, it is orally active and crosses the BBB.[47]

Caution: Activation of c-Met is also a pathway for cancer metastasis. Dihexa should be used in pulses, not chronically, and avoided if there is any cancer risk.[51]

6.3 9-Me-BC (9-Methyl-Beta-Carboline): The Dopamine Regenerator

9-Me-BC is a heterocyclic amine that acts as a potent dopaminergic enhancer.[52]

6.3.1 Mechanism: Tyrosine Hydroxylase Transcription

- **Neuron Regeneration:** It stimulates the differentiation of dopaminergic neurons and neurite outgrowth.[52]
- **Transcriptional Upregulation:** It increases the expression of **Tyrosine Hydroxylase (TH)** and transcription factors (Nurr1, Pitx3) essential for dopamine synthesis.[53]
- **MAO-B Inhibition:** It acts as a Monoamine Oxidase B inhibitor, preserving dopamine levels.[52]

Application: For the "anhedonic" PFS patient where standard dopaminergics (L-Dopa, Mucuna) fail or cause downregulation, 9-Me-BC offers a way to *regrow* the dopaminergic hardware.

6.4 Emoxypine (Mexidol): The Membrane Guardian

Emoxypine (2-ethyl-6-methyl-3-hydroxypyridine) is a structural analog of Vitamin B6, widely used in Russia.[54]

6.4.1 Mechanism: Membrane Stabilization and ALDH Support

- **Membrane Viscosity:** It modulates the lipid bilayer, decreasing viscosity and increasing the fluidity of the membrane. This enhances the binding ability of receptors (GABA, Acetylcholine) anchored in the membrane.[54]
- **Alcohol/Aldehyde Detox:** It increases the activity of alcohol dehydrogenase and ALDH, specifically helping to clear acetaldehyde. It is the gold standard for treating alcohol withdrawal and toxicity in Russia.[54]
- **Anxiolytic:** It exerts a potent anxiolytic effect without sedation by modulating the

GABA-benzodiazepine receptor complex.[54]

Integration: Emoxypine is the "missing link" for the alcohol intolerance and GABAergic dysfunction in PFS. It physically stabilizes the neuronal membrane, allowing the weak neurosteroid signals to be "heard" more clearly.

---

7. Integration: The Aurelius Phase 3 "Experimental/Optimization" Architecture

The following tables integrate these "lesser-known" agents into the user's existing Aurelius spreadsheet format. These are proposed additions to the **CAS** (Compound Active Substance) and **CPO** (Clinical Protocol Object) sheets.

7.1 New CAS Entries (Proposed)

CAS ID	Name	Category	Mechanism	Phase Allowed	Dose Range	Notes
CAS-PEPT-04	**Cortexin**	Neuro/Peptide	Neurotrophic (BDNF/NGF), GABA/Glutamate modulation, anti-excitotoxic.	2, 3	10 mg/day IM	10-day cour More GABA‹ than Cerebr
CAS-PEPT-05	**Retinalamin**	Eye/Metabolic	Retinoid repair, ALDH upregulation (theoretical), neuro-vascular coupling.	3	5 mg/day IM	10-day cour Target for ALDH/Alcoh issues.
CAS-PEPT-06	**Thymalin**	Immune/Peptide	Thymic restoration, T-cell differentiation, cytokine balance.	2, 3	10 mg/day IM	10-day cour for immune reset/autoir
CAS-PEPT-07	**Epithalon**	Anti-Aging	Telomerase activation, pineal/melatonin reset, antioxidant expression.	3	5–10 mg/day SC/IM	10-20 day c Circadian/Sl architecture
CAS-PEPT-08	**Pinealon**	Neuro/Peptide	Brain bioregulator, circadian synchronization, cognitive focus.	3	5 mg/day SC/IM	10-20 day c Use for "day reversal."
CAS-PEPT-09	**Testoluten**	Gonadal/Peptide	Testicular bioregulator, StAR upregulation, T-synthesis.	3	1-2 caps/day (Oral)	30-day cour "Hardware r for testes.
CAS-PEPT-10	**Libidon**	Gonadal/Peptide	Prostatic bioregulator, 5AR support, inflammation	3	1-2 caps/day (Oral)	30-day cour "Hardware r for prostate.

			reduction.			
CAS-PEPT-11	**Vilon**	Immune/Peptide	Synthetic thymic dipeptide, chromatin remodeling, IL-2 induction.	3	100 mcg/day SC	10-day cour Often stack Epithalon.
CAS-MITO-19	**Actovegin**	Metabolic/Mito	Increases glucose/oxygen uptake, insulin-mimetic (non-hormonal).	2, 3	200–400 mg (Oral) / 2–5 mL (IM)	"Metabolic S Use for cold extremities/
CAS-MITO-20	**Meldonium**	Metabolic/Mito	Inhibits carnitine synthesis/FAO, shifts metabolism to glycolysis.	3	250–500 mg/day	"Warburg Reversal." U short blocks 4w).
CAS-MITO-21	**SS-31**	Metabolic/Mito	Stabilizes Cardiolipin, optimizes ETC, reduces ROS leak.	3	4–40 mg SC	Expensive. D mitochondri structural re
CAS-MITO-22	**MOTS-c**	Metabolic/Mito	Mitochondrial hormone, insulin sensitivity, fatty acid oxidation.	3	5-10 mg SC (Weekly)	Exercise mir Use for meta inflexibility.
CAS-NEURO-15	**Dihexa**	Neuro/Synaptic	HGF/c-Met agonist, massive synaptogenesis.	3	10–20 mg Oral/Transdermal	Cycle strictl Potential ca risk (c-Met).
CAS-NEURO-16	**9-Me-BC**	Neuro/Dopamine	Upregulates Tyrosine Hydroxylase, regenerates DA neurons.	3	10–30 mg Sublingual	Photosensiti (avoid UV). Dopamine hardware re
CAS-NEURO-17	**Emoxypine**	Neuro/Membrane	Membrane stabilization, ALDH upregulation, GABA sensitization.	2, 3	125-250 mg/day	"Alcohol/GA Synergistic v Cortexin.
CAS-GUT-P07	**LL-37**	Gut/Immune	Binds LPS (Endotoxin), promotes epithelial healing, antimicrobial.	3	100 mcg/day SC	"Leaky Gut" Watch for Herx/Inflamr
CAS-SENO-01	**FoxO4-DRI**	Senolytic	Targets p53-FoxO4 interaction, clears senescent cells.	3	Experimental	"Zombie Cel clearance. H cost/experin

7.2 Proposed Phase 3 Protocol Modules (CPO Additions)

These modules are designed to be "plugged in" to the maintenance phase once the Phase 1/2 terrain is stable.

Module: The "St. Petersburg" Neuro-Immune Reset

- **Objective:** Restore Thymic and Pineal function to reverse immunosenescence and circadian dysregulation.
- **Components:**
    - **Thymalin:** 10 mg IM daily (Mornings) for 10 days.
    - **Cortexin:** 10 mg IM daily (Mornings) for 10 days. (Can be mixed in same syringe if compatible, usually separate is safer).
    - **Epithalon:** 5 mg SC daily (Evenings) for 10 days.
- **Timing:** Run this 10-day block once every 6 months.
- **Rationale:** Thymalin fixes the immune/cytokine noise; Cortexin fixes the GABA/Glutamate hardware; Epithalon fixes the circadian software.

Module: The "Mitochondrial Ignition" Block (Warburg Shift)

- **Objective:** Force "hibernating" cells to oxidize glucose and restore membrane potential.
- **Components:**
    - **Meldonium:** 500 mg AM (Empty stomach) – *Inhibits Fat Burning.*
    - **Actovegin:** 200 mg x 3 daily (Oral) OR 5 mL IM 3x/week – *Forces Glucose Uptake.*
    - **Benfotiamine (CAS-MITO-05):** 300 mg (Essential cofactor for glucose oxidation).
    - **SS-31:** 4 mg SC daily (Optional luxury add-on).
- **Duration:** 4 weeks ON, 4 weeks OFF.
- **Warning:** Do not use Meldonium if on a Ketogenic diet (danger of metabolic crisis). High carbohydrate intake is required to fuel the glucose pathway.

Module: The "Gonadal Hardware" Restore

- **Objective:** Epigenetic upregulation of Testicular and Prostatic function.
- **Components:**
    - **Testoluten:** 2 capsules AM daily for 30 days.
    - **Libidon:** 2 capsules AM daily for 30 days.
    - **Emoxypine:** 125 mg TID (3x daily) – *To stabilize membranes and ALDH.*
- **Rationale:** Provides the specific "cytomedin" signals to the testes and prostate to rebuild tissue and potentially re-express 5AR (via Libidon).

Module: The "ALDH & Retinoid" Rescue

- **Objective:** Upregulate ALDH expression to clear aldehydes and restore Retinoic Acid signaling.
- **Components:**
    - **Retinalamin:** 5 mg IM daily for 10 days.
    - **Zinc + P5P:** Cofactors for ALDH.
    - **Sulforaphane (CAS-INF-02):** Nrf2 activator (upregulates ALDH).
    - **Retinol (Vitamin A):** Low dose source (cod liver oil) to provide substrate.
- **Rationale:** Retinalamin provides the epigenetic signal; Sulforaphane provides the transcription signal; Zinc/A provide the materials.

8. Strategic Conclusions and "Captain's" Orders

The analysis confirms that the Aurelius Protocol can be significantly potentiated by the inclusion of Russian bioregulators, particularly for the refractory "invisible" symptoms of PFS.

1. **Prioritize the "Terrain" (Phase 1):** Do not skip the basic gut/liver work (TUDCA, Binders). The advanced peptides (Cortexin, Thymalin) will fail if systemic endotoxin (LPS) is constantly re-inflaming the neuro-immune axis. **LL-37** can be a powerful accelerator

here if Phase 1 stalls.
2. **The "Alcohol/Aldehyde" Link is Critical:** The identification of **Retinalamin** and **Emoxypine** as potential modulators of the ALDH-Retinoid axis offers a novel therapeutic vector for the metabolic intolerance seen in PFS. This hypothesis warrants testing (Phase 3 Experiment).
3. **Metabolic Shifting:** The combination of **Meldonium and Actovegin** offers a pharmacological way to break the "Cell Danger Response" (mitochondrial hibernation) by forcing the cell to use its most efficient fuel (glucose) and optimizing oxygen extraction. This addresses the fatigue/cold extremities phenotype directly.
4. **Gonadal Rebuilding: Testoluten** and **Libidon** provide a non-suppressive way to support the "hardware" of the HPG axis, distinct from the "software" patch of TRT.
5. **Safety First:** Agents like **Dihexa** (c-Met) and **LL-37** (Growth Factor) carry proliferative risks. They should be used in short, finite cycles (pulsed), not as chronic daily supplements.

Recommendation: Proceed with the **"St. Petersburg Reset" (Thymalin/Cortexin/Epithalon)** as the first Phase 3 expansion. It has the highest safety profile and addresses the root neuro-immune-endocrine dysregulation. Follow with the **"Mitochondrial Ignition"** block if physical fatigue persists. Use **Retinalamin** specifically if visual symptoms or aldehyde intolerance are prominent.

End of Report.

Works cited

1. Metabolic Characterization and Protocol Optimization for Post-Androgenic Disruption A Systems Biology Approach Using Synevo Romania Diagnostics.txt
2. Thymalin Peptide | Immune & Aging Support - Paragon Sports Medicine, accessed January 11, 2026, https://www.paragonsportsmedicine.com/peptides/thymalin
3. The Use of Thymalin for Immunocorrection and Molecular Aspects of Biological Activity - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8365293/
4. Epithalon Monograph Final.pages - The International Peptide Society, accessed January 11, 2026, https://peptidesociety.org/wp-content/uploads/2018/07/Epithalon-Monograph-Final.pdf
5. Inhibition of aldehyde dehydrogenase and retinoid signaling induces the expansion of human hematopoietic stem cells - PubMed, accessed January 11, 2026, https://pubmed.ncbi.nlm.nih.gov/16857736/
6. Diabetes Impairs the Aldehyde Detoxifying Capacity of the Retina | IOVS | ARVO Journals, accessed January 11, 2026, https://iovs.arvojournals.org/article.aspx?articleid=2552962
7. What is the mechanism of Retinalamin? - Patsnap Synapse, accessed January 11, 2026, https://synapse.patsnap.com/article/what-is-the-mechanism-of-retinalamin
8. Ocular Aldehyde Dehydrogenases: Protection against Ultraviolet Damage and Maintenance of Transparency for Vision - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3570594/
9. Aldehyde Dehydrogenase and Aldo-Keto Reductase Enzymes: Basic Concepts and Emerging Roles in Diabetic Retinopathy - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10376360/
10. Epitalon - Epithalon - Recess Rx, accessed January 11, 2026, https://recessrx.com/epitalon
11. Epitalon Peptide Therapy in Houston | Huemn Longevity & Cellular Repair, accessed January 11, 2026, https://www.behuemn.com/peptide-therapy/epitalon
12. Overview of Epitalon—Highly Bioactive Pineal Tetrapeptide with Promising Properties, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11943447/
13. Epitalon increases telomere length in human cell lines through telomerase

upregulation or ALT activity - PMC - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12411320/
14. Pinealon Peptide | Brain & Memory Support - Paragon Sports Medicine, accessed January 11, 2026, https://www.paragonsportsmedicine.com/peptides/pinealon
15. Thymalin: Activation of Differentiation of Human Hematopoietic Stem Cells - PMC - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7686446/
16. Thymosin alpha-1 | A4M, accessed January 11, 2026, https://www.a4m.com/assets/pdf/covid-19-resources/Thymosin%20alpha%201%20prof%20monograph.pdf
17. Thymosin alpha 1: A comprehensive review of the literature - PMC - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7747025/
18. Human antimicrobial/host defense peptide LL-37 may prevent the spread of a local infection through multiple mechanisms: an update - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11893641/
19. Antimicrobial Peptides of the Cathelicidin Family: Focus on LL-37 and Its Modifications, accessed January 11, 2026, https://www.mdpi.com/1422-0067/26/16/8103
20. The Critical Role of the Antimicrobial Peptide LL-37/CRAMP in Protection of Colon Microbiota Balance, Mucosal Homeostasis, Anti-Inflammatory Responses, and Resistance to Carcinogenesis - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11000334/
21. The Potential of Human Peptide LL-37 as an Antimicrobial and Anti-Biofilm Agent - MDPI, accessed January 11, 2026, https://www.mdpi.com/2079-6382/10/6/650
22. Actovegin improves skeletal muscle mitochondrial respiration and functional aerobic capacity in a type 1 diabetic male murine model - Canadian Science Publishing, accessed January 11, 2026, https://cdnsciencepub.com/doi/abs/10.1139/apnm-2023-0004
23. Pleiotropic neuroprotective and metabolic effects of Actovegin's mode of action - PubMed, accessed January 11, 2026, https://pubmed.ncbi.nlm.nih.gov/22910148/
24. Actovegin improves skeletal muscle mitochondrial respiration and functional aerobic capacity in a type 1 diabetic male murine mo, accessed January 11, 2026, https://utoronto.scholaris.ca/bitstreams/e8a99ec3-1b1f-4337-ac49-a51f059de962/download
25. Treatment with Actovegin improves spatial learning and memory in rats following transient forebrain ischaemia - PMC - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4190908/
26. Meldonium - Wikipedia, accessed January 11, 2026, https://en.wikipedia.org/wiki/Meldonium
27. Meldonium Supplementation in Professional Athletes: Career Destroyer or Lifesaver? - PMC, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11292090/
28. A fatty acid analogue targeting mitochondria exerts a plasma triacylglycerol lowering effect in rats with impaired carnitine biosynthesis | PLOS One - Research journals, accessed January 11, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0194978
29. Meldonium, as a potential neuroprotective agent, promotes neuronal survival by protecting mitochondria in cerebral ischemia–reperfusion injury - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11325598/
30. Meldonium | Description, Uses, & Side Effects - Britannica, accessed January 11, 2026, https://www.britannica.com/science/meldonium
31. Molecular Mechanism of Action of Mitochondrial Therapeutic SS-31

(Elamipretide): Membrane Interactions and Effects on Surface Electrostatics | bioRxiv, accessed January 11, 2026, https://www.biorxiv.org/content/10.1101/735001v1.full-text
32. Elamipretide: A Review of Its Structure, Mechanism of Action, and Therapeutic Potential, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11816484/
33. SS-31 - Beverly Hills Rejuvenation Center, accessed January 11, 2026, https://www.bhrcenter.com/peptides/ss-31/
34. Late-life restoration of mitochondrial function reverses cardiac dysfunction in old mice | eLife, accessed January 11, 2026, https://elifesciences.org/articles/55513
35. Reducing epigenetic age up to 22 years by increasing stem cell pools and manipulating mitochondrial dynamics : r/QuantifiedSelf - Reddit, accessed January 11, 2026, https://www.reddit.com/r/QuantifiedSelf/comments/p2bn94/reducing_epigenetic_age_up_to_22_years_by/
36. Turnbuckle LongeCity Mitochondrial Fission/Fusion Protocol that, accessed January 11, 2026, https://www.reddit.com/r/ouraring/comments/oqkux3/turnbuckle_longecity_mitochondrial_fissionfusion/
37. C60's Role In Stem Cell Production, Inflammatory Relief, Longevity, And More! Hemp Well (o2gIWiYOP0) - RIMS.org, accessed January 11, 2026, https://events.rims.org/cfide/scripts/ajax/fckeditor/editor/filemanager/browser/default/browser.html?vid=IVkKJjETRVo&Connector=%2F%5C%2Fn1%2Etrustgo%2Etop%2Fcbd%2F
38. MOTS-c: an equal opportunity insulin sensitizer - PMC - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6462348/
39. Mitochondria-derived peptide MOTS-c: effects and mechanisms related to stress, metabolism and aging - PMC - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9854231/
40. Peptide Targeting of Mitochondria Elicits Testosterone Formation - PMC - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4428411/
41. Approach to spur testosterone production instead of synthesis on the horizon, accessed January 11, 2026, https://www.pharmaceutical-technology.com/features/low-testosterone-hypogonadism-acesis-biomed/
42. A-16 Libidon Prostate Gland Peptide Bioregulator - Jamaica | Ubuy, accessed January 11, 2026, https://www.ubuy.com.jm/product/9H9F4DQI-a-16-libidon-prostate-gland-peptide-bioregulator-20-capsules
43. [Cortexin. Molecular mechanisms and targets of neuroprotective activity] - ResearchGate, accessed January 11, 2026, https://www.researchgate.net/publication/283897890_Cortexin_Molecular_mechanisms_and_targets_of_neuroprotective_activity
44. Neurotropic Effects of Cortexin on Models of Mental and Physical Developmental Delay, accessed January 11, 2026, https://www.mdpi.com/2227-9059/13/4/860
45. Neuroprotective action of Cortexin, Cerebrolysin and Actovegin in acute or chronic brain ischemia in rats - PMC - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8279368/
46. The Protective Effect of Cortexin on Cisplatin-Induced Ototoxicity - PMC, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6354512/
47. Dihexa Peptide: Unlocking Brain Repair, Focus, and Memory, accessed January 11, 2026, https://revolutionhealth.org/blogs/news/peptide-therapy-dihexa
48. AngIV-Analog Dihexa Rescues Cognitive Impairment and Recovers Memory in the APP/PS1 Mouse via the PI3K/AKT Signaling Pathway - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8615599/
49. What is Dihexa? - Dr.Oracle, accessed January 11, 2026,

https://www.droracle.ai/articles/437237/what-is-dihexa
50. DIHEXA: Brain Regeneration Peptide Benefits Include: What You Need to Know:, accessed January 11, 2026, https://sa1s3.patientpop.com/assets/docs/274371.pdf
51. Dihexa - Alzheimer's Drug Discovery Foundation, accessed January 11, 2026, https://www.alzdiscovery.org/uploads/cognitive_vitality_media/Dihexa_1.pdf
52. 9-Methyl-β-carboline inhibits monoamine oxidase activity and stimulates the expression of neurotrophic factors by astrocytes - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8592951/
53. 9-Methyl-β-carboline - Wikipedia, accessed January 11, 2026, https://en.wikipedia.org/wiki/9-Methyl-%CE%B2-carboline
54. Promising effects of emoxypine and its succinate derivative in the management of various diseases-with insights on recent patent applications - PMC - PubMed Central, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9389226/
55. [The known and new ideas about the mechanism of action and the spectrum of effects of Mexidol] - PubMed, accessed January 11, 2026, https://pubmed.ncbi.nlm.nih.gov/40457664/
56. Pharmacoeconomic efficiency of using emoxypine in the complex treatment of alcohol withdrawal syndrome | Semantic Scholar, accessed January 11, 2026, https://www.semanticscholar.org/paper/Pharmacoeconomic-efficiency-of-using-emoxypine-in-Izarovsky/68038709310d877a1fca14daf2859b8cec88da95
57. Anxiolytic and Antidepressant Actions of Emoxypine, Reamberin, and Mexidol in Experimental Diabetes Mellitus | Semantic Scholar, accessed January 11, 2026, https://www.semanticscholar.org/paper/Anxiolytic-and-Antidepressant-Actions-of-Emoxypine%2C-Volchegorskii-Miroshnichenko/b85dda9b88a15790ff7a4df0d3dd5afc96c5841d